

**24**  
**L2**

Solution: **A**

For an alternating supply,

$$\langle P \rangle = P$$

$$P = \frac{V_{rms}^2}{R} \dots\dots(1)$$

Across the same load, to have average power to be 3P,

$$\langle P \rangle_{new} = \frac{V_{rms\ new}^2}{R}$$

$$3P = \frac{V_{rms\ new}^2}{R} \dots(2)$$

$$(1) / (2)$$

$$1/3 = \frac{\frac{V_{rms}^2}{R}}{\frac{V_{rms\ new}^2}{R}}$$

$$V_{rms, new} = \sqrt{3V_{rms}^2} = \sqrt{3(4)^2} = 6.9\text{ V}$$

**25**

Solution: **A**